

Xiaowen (Winn) Ge

RESEARCHER IN AI & XR
APPLICATIONS FOR MUSIC
EDUCATION

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Methods & Tools

Qualitative: semi-structured / video-elicited interviews, observation, thematic analysis, NVivo, PRISMA scoping reviews.

Computational: Python (librosa for music-feature extraction).

Languages

Mandarin (native), English (fluent).

Musical Practice

20 years' piano training; 13 years' vocal training in traditional Chinese folk song singing.

Conferences & Presentations

ISATT 2025, Glasgow

Quality and Aesthetics in Music Teachers' Perspectives

LKALE SIG / BAAL 2024

Exploring Multimodal Communication in Music Education

RiME 2023

Chinese Music Teachers' Perceptions of Fostering Creativity (Conference Proceedings); An Empirical Model of Creativity in Music Education (Online presentation)

UofG PhD-led Webinar, 2020

Using Digital Tools to Bridge the Distance Divide

Profile

PhD in Education (University of Glasgow, 2024), qualitative research design specialist (interviews, video-recorded observation, thematic analysis). Current work moves into AI/XR music-education contexts: designing Syn-Aesthetica (a Python/C++ system for music-feature analysis and emotion classification, intended to drive physical/kinetic output) and teaching NVivo-based qualitative methods to VR/generative-AI pedagogy postgraduates at Liverpool. The next step is applying that same qualitative approach to how existing AI/XR tools are actually used, understood and negotiated by learners and teachers — evaluation and methodology, not tool-building for its own sake.

Research Focus

Creativity in education — core PhD strand applying Csikszentmihalyi's Domain-Field-Individual model — the Field as institutional gatekeeper selecting which novelty is transmitted — to how teachers mediate three curriculum-defined creativity dimensions (sound exploration, improvisation, composition); related peer-reviewed journal article forthcoming (*Music Education Research*).

Cross-stage teacher mediation of musical engagement — collaborative secondary analysis combining two independent PhD datasets (with Dr Sicong Li and Prof. Oscar Odena) comparing nursery and primary music education in China; manuscript in development, target: *Australasian Journal of Early Childhood*.

Related projects

Syn-Aesthetica (2025–present, with jewellery-design PhD collaborator Dong Ding, Edinburgh College of Art) — Python/C++ tool intended to combine librosa feature extraction with a self-trained lightweight emotion classifier (public affective-music datasets) to drive physical/kinetic output via Ding's sculptural mechanisms. Role: project strategy and funding-proposal writing, analysis pipeline, classifier training, and the educational-evaluation framework; prototype/demo stage. Two outputs in preparation: a scoping review positioning the field (Ding first author; target: *CoDesign*) and a software paper (Ge first author; target: *JOSS*).

XR research design for inclusive pedagogy — authored a full proposal for the ESRC North West Social Science DTP (NWSSDTP) Postdoctoral Fellowship (October 2025 entry) on XR-based music pedagogy for neurodivergent learners, drawing on AR/holographic display technology from Liverpool's Design4Life lab (Dr Hao-Ming Xia's museum and cultural-heritage AR research) — not taken forward at the institutional nomination stage; developed a participatory co-design and cognitive-science evaluation framework now informing the evaluation design being developed for Syn-Aesthetica.

Service

Reviewer, Learning Environments Research (Springer Nature).

Teaching Experience

Guest Lecturer, University of Liverpool

2024–present

Design and deliver standalone workshops on scoping reviews, thematic analysis (NVivo) and publication pathways for postgraduates in VR/generative-AI pedagogy and digital humanities; invited back each cohort following direct faculty feedback that the applied focus of the teaching visibly increased students' engagement and equipped them to apply the methods in their own dissertations.

Workshops and guest teaching reach approximately 60 undergraduate and postgraduate students to date, including method-specific teaching for postgraduate researchers (5 PhD) covering question framing, ethics applications, fieldwork design and write-up.

Freelance Museum Educator, National Museums Scotland; Hunterian Museum, University of Glasgow

2018–present

Research, write and deliver interpretation for school, university, public and international audiences; train and assess new guides.

Education

PhD in Education — University of Glasgow

01/2019–06/2024

Thesis: *Music teachers' perceptions of creativity in the context of twenty-first century Chinese education* (viva 27 September 2023). Supervisors: Prof. Oscar Odena, Prof. Kay Livingston.

Qualitative design: 17 semi-structured interviews, 24 lesson observations, critical-incident charting, researcher diary; thematic analysis in NVivo. Produced an original seven-stage model of creativity development in music education. Key finding: conservatoire-trained instrumentalists received formal improvisation training; generalist teachers engaged with it only informally.

MSc in Education (Research) — University of Edinburgh

09/2017–08/2018

Dissertation: *The Role of Higher Education in Youth Transition to Employment*. Mixed methods, NVivo, SPSS & Python.

BA in English (Distinction) — Shandong First Medical University

09/2013–06/2017

Dissertation: *The Application of Task-based Language Teaching in IELTS Teaching*. Undergraduate research project (project lead, 4-person team): *An Analysis of Common English Learning Methods and Their Uses* — funded by a Taishan Medical University internal research grant (¥3,000).

Publications

PUBLISHED

Ge, X., Odena, O., & Livingston, K. (2026). Teachers' Perceptions of Creativity: A Practice-Informed Model from Primary School Music and Piano Teaching in China. *Research Studies in Music Education*.
<https://doi.org/10.1177/1321103X261460254>

UNDER REVIEW

Ge, X., Odena, O., & Livingston, K. Creativity in Music Education: The Influence of Policy and Experience on Piano and School Music Teachers' Perceptions in Shandong, China. *Music Education Research*.

IN PREPARATION

Li, S., Ge, X., & Odena, O. Teacher mediation of musical engagement across early childhood and primary music education in China: a qualitative secondary analysis. Manuscript in development; target: *Australasian Journal of Early Childhood*.

Ding, D. & Ge, X. Physical and Tangible Representations of Music: A Scoping Review of Design and Mapping Approaches and Their Applications in Learning, Wellbeing and Creative Practice. *CoDesign*.

Ge, X. & Ding, D. Syn-Aesthetica: an open-source tool mapping music analysis to physical/kinetic representation. *Journal of Open Source Software*.